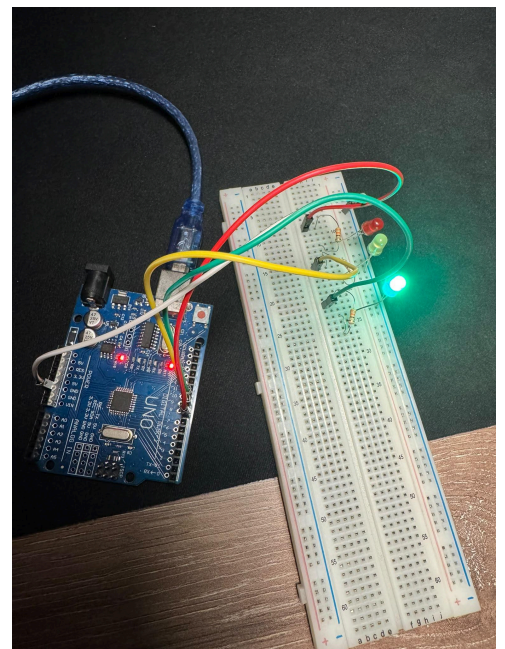
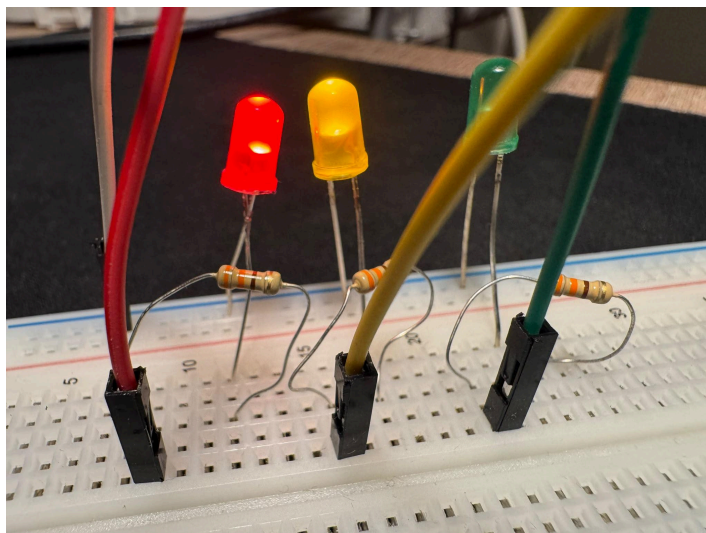
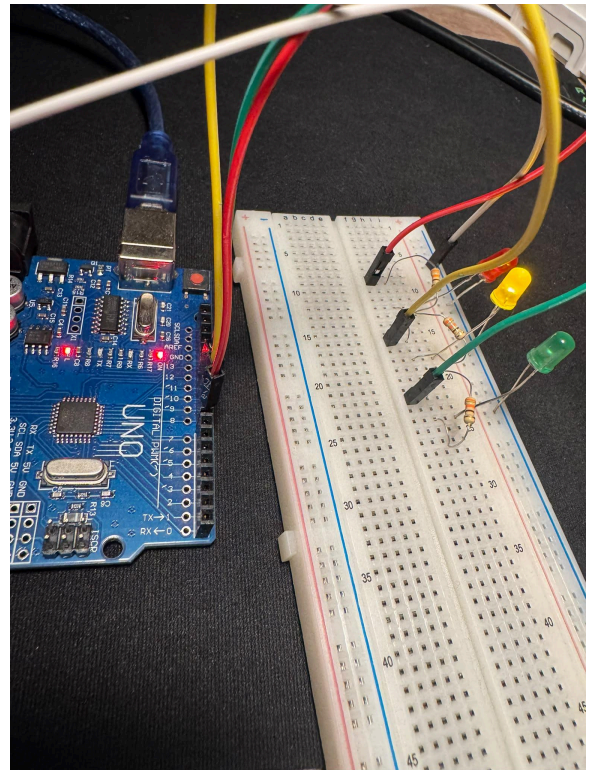
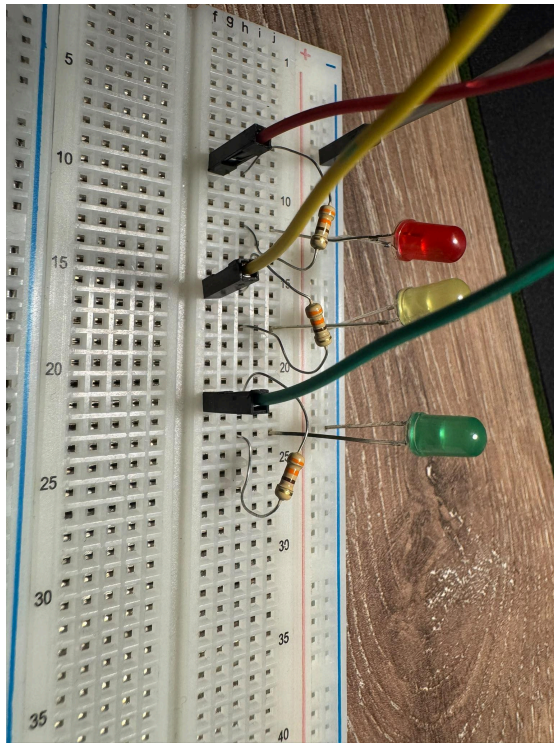


Arduino Traffic Light



Circuit Components

- LED (light-emitting diode)

LEDs typically operate at around 15mA (milliamperes). Applying 5V directly may damage the LED, so it must be protected with a resistor.

Ohm's Law: $V = I \times R$

$V \rightarrow$ Voltage (Volt (V))

$I \rightarrow$ Current (Ampere (A))

$R \rightarrow$ Resistance (Ohm (Ω))

- Resistor

LEDs cannot tolerate excessive current. Therefore, each LED is connected in series with a 330 Ω resistor to limit the current and protect the LED.

$$R = (V_s - V_f) / I$$

- Breadboard

A breadboard is a reusable plastic block filled with small holes used to build and test electronic circuits without soldering.

- Arduino

A printed circuit board that houses the microcontroller chip alongside extra supportive parts like power regulators, USB communication interfaces and easy-to-use pin headers.

- Jumper Wires

Jumper wires are short electrical wires with a pin or connector at each end, used to join two points in a circuit without soldering.

Circuit Connection

Red LED

Connect the short leg (cathode, -) directly to the breadboard GND rail.

Connect the long leg (anode, +) to Arduino digital pin 8 through a 330 Ω resistor.

Yellow LED

Connect the short leg (cathode, -) directly to the breadboard GND rail.

Connect the long leg (anode, +) to Arduino digital pin 9 through a 330 Ω resistor.

Green LED

Connect the short leg (cathode, -) directly to the breadboard GND rail.

Connect the long leg (anode, +) to Arduino digital pin 10 through a 330Ω resistor.

How It Works

The Arduino simulates a basic traffic light using three LEDs.

The traffic light follows this sequence:

- 1- The red LED turns on, indicating Stop.
- 2- The red and yellow LEDs turn on together, indicating Prepare to Go.
- 3- The green LED turns on, indicating Go.
- 4- The yellow LED turns on, indicating Prepare to Stop.
- 5- The sequence repeats continuously.

Traffic Light Sequence

Step 1 - Stop

Pin 8 → 330Ω Resistor → Red LED (Anode) → Red LED (Cathode) → GND

Pin 9 = LOW (Yellow LED OFF)

Pin 10 = LOW (Green LED OFF)

Step 2 - Prepare to Go

Pin 8 → 330Ω Resistor → Red LED (Anode) → Red LED (Cathode) → GND

Pin 9 → 330Ω Resistor → Yellow LED (Anode) → Yellow LED (Cathode) → GND

Pin 10 = LOW (Green LED OFF)

Step 3 - Go

Pin 10 → 330Ω Resistor → Green LED (Anode) → Green LED (Cathode) → GND

Pin 8 = LOW (Red LED OFF)

Pin 9 = LOW (Yellow LED OFF)

Step 4 - Prepare to Stop

Pin 9 → 330Ω Resistor → Yellow LED (Anode) → Yellow LED (Cathode) → GND

Pin 8 = LOW (Red LED OFF)

Pin 10 = LOW (Green LED OFF)

Code

```
#define red_led 8

#define yellow_led 9

#define green_led 10

void setup() {

    // Set pin 8, 9 and 10 as a digital output.

    pinMode(red_led, OUTPUT);

    pinMode(yellow_led, OUTPUT);

    pinMode(green_led, OUTPUT);

}
```

```
void loop() {

    digitalWrite(red_led, HIGH);

    digitalWrite(yellow_led, LOW);

    digitalWrite(green_led, LOW);

    delay(3000);

    digitalWrite(red_led, HIGH);

    digitalWrite(yellow_led, HIGH);

    digitalWrite(green_led, LOW);

    delay(1000);

    digitalWrite(red_led, LOW);

    digitalWrite(yellow_led, LOW);

    digitalWrite(green_led, HIGH);

    delay(2000);

    digitalWrite(red_led, LOW);

    digitalWrite(yellow_led, HIGH);

    digitalWrite(green_led, LOW);

    delay(1000);

}
```